

Thikana

An AI-Native Bilingual Rental Marketplace for Bangladesh

Team: IUT_TwinCoder

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Problem Statement

According to University Grants Commission (UGC) data, nearly **60% of public university students in Bangladesh lack access to university residential facilities** and must rely entirely on private rentals. For students relocating to Dhaka from other districts, finding trustworthy, affordable accommodation remains a severe challenge. With no centralised platform, they physically scour neighbourhoods by vehicle or rickshaw, spending days searching for vacant rooms based on handwritten signs and local gossip. A preliminary survey of 50 students across IUT, DIU, and DUET confirmed this: **84% relied solely on Facebook groups** to find housing, while **71% reported encountering misleading or fake listings** during their search.

The problem is equally acute on the supply side. Landlords have no tool to write effective bilingual listings, no benchmark for a fair rent, and no organised dashboard to track which tenants have paid and which have not. Students sharing accommodation also face a daily arithmetic burden: splitting rent, calculating shared meal contributions, dividing utility bills, and estimating whether their monthly allowance will last. For young professionals and families, the same opacity applies — the absence of a centralised, verifiable platform turns every rental search into a guessing game for all parties.

Proposed Solution

Thikana is a full-stack rental marketplace that brings landlords, student renters, young professionals, and family tenants onto one bilingual (Bangla/English), mobile-first platform. AI is embedded deliberately and only where it delivers genuine reasoning capability — natural-language understanding, anomaly detection, and generative output. Routine platform functions such as payment tracking, bill splitting, distance calculation, and photo uploading are handled by standard software without invoking an LLM.

The platform serves two distinct rental contexts under one roof: *student housing*, with tools for flatmate matching, daily meal-cost splitting, and university-distance filtering; and *family and professional housing*, with broader search filters, verified

landlord profiles, and neighbourhood safety insights.

Key Features

AI-Powered Features

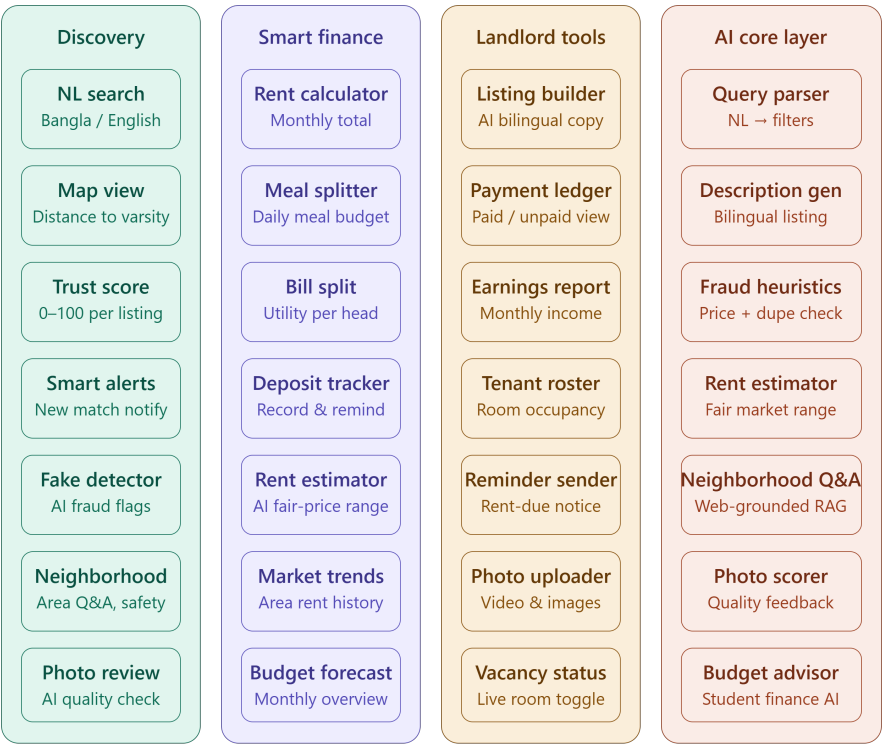
(require language reasoning or anomaly detection)

- **Natural-language search** — Parses mixed Bangla/English queries (e.g. “Mirpure 2 room 12k er niche”) into structured database filters instantly.
- **AI listing generator** — Converts a landlord’s structured form inputs into a polished, bilingual listing description in one click.
- **AI integrity score (0–100)** — A rule-based and ML hybrid system: rental price distributions flag below-market anomalies; TF-IDF cosine similarity detects duplicate descriptions; perceptual hashing (pHash) identifies reused or stock photos.
- **Neighbourhood assistant** — Answers area-specific questions (e.g. “Is this safe for families?”) using LLM reasoning grounded in real-time web search, reducing hallucination risk.
- **Smart rent estimator** — Suggests a fair rent range derived from local price distributions for the given area, size, floor, and furnishing level.

Platform Features

(standard software, no LLM)

- **University-distance map view** — Geo-anchored listings show calculated travel time from a chosen university gate.
- **Landlord payment dashboard** — Tracks tenant payment status, sends automated rent-due reminders, and generates monthly earnings reports.
- **Student finance toolkit** — Splits shared rent, divides utility bills, calculates daily meal-cost contributions, and forecasts monthly budget sufficiency.
- **Flatmate matchmaking** — Matches students by university, budget, and lifestyle preferences.
- **Smart match alerts** — Notifies tenants when a listing matching their saved criteria goes live.
- **Photo quality scorer** — Rates uploaded listing photos and prompts landlords to improve them before publishing.



Innovation and Uniqueness

No existing platform in Bangladesh combines geo-anchored listings with a university-distance filter, a bilingual AI listing assistant, a student-specific financial toolkit, and a landlord payment ledger under one roof. **Thikana**’s innovation lies not in any single feature but in the integration: a platform designed around how Bangladeshi renters — particularly students arriving from outside Dhaka — actually search, communicate, and manage money.

Students operate differently from professionals: they share meals, split costs among three to six people, receive irregular allowances from families, and prioritise proximity to university gates. **Thikana** treats these as first-class design constraints rather than edge cases.

Existing Solution	Key Limitation
Facebook Groups	No verification, no search filters, no trust signals
Bikroy.com	General classifieds; not student-focused; no AI features
Bproperty	Targets large/commercial properties; inaccessible price range
Informal Brokers	High commission, low transparency, no digital record

Effective and Ethical Use of AI

AI in **Thikana** is applied only at the five touchpoints where rule-based logic is insufficient. Each touchpoint is explainable, documented, and defensible:

Touchpoint	Function and Method
Generation	LLM converts structured landlord inputs into bilingual listing copy (Bangla and English).
Extraction	LLM parses free-text tenant queries into structured database filters (area, rooms, rent, preferences).
Integrity score	Rule-based + ML hybrid: price distribution anomaly detection, TF-IDF duplicate description check, and pHash photo reuse detection. Outputs a transparent 0–100 score on every listing card.
Grounded retrieval	LLM reasons over live web search results to answer neighbourhood queries, minimising hallucination.
Budget advisory	Rule-based threshold check warns students when a chosen rent exceeds a safe share of their stated monthly allowance; no LLM required.

Every prompt, input schema, and fallback path is documented and understood by the development team, ensuring that each AI decision can be explained and defended.

Feasibility and Real-World Applicability

Production-Ready Architecture & Tech Stack:

- **Frontend & UI:** **Next.js (App Router)** utilizing Server-Side Rendering (SSR) for optimal mobile performance and SEO, styled with a mobile-first **Tailwind CSS** layout.
- **Backend-as-a-Service:** **Supabase** handles secure JWT-based authentication, PostgreSQL relational data for rapid multi-parameter listing filters, and Storage buckets for property media.
- **AI & Intelligence Layer:** **Google Gemini API** (orchestrated server-side via Edge Functions) powers the natural-language search parser, RAG-driven neighborhood Q&A, and automated bilingual translation.
- **Geospatial Engine:** **Mapbox API** combined with **Leaflet.js** provides lightweight, highly performant interactive mapping, exact address geo-anchoring, and custom campus-distance calculations.

The two-sided market has clear incentives on both sides: tenants want to find verified rooms faster, and landlords want to fill vacancies and get paid on time. A

focused soft-launch strategy — partnering with landlords in two to three university-adjacent areas of Dhaka before public release — addresses the cold-start problem. Scout and building-manager accounts accommodate landlords who are less comfortable with direct smartphone-based listing management.

Future Potential and Scalability

The platform's architecture supports expansion along three axes. *Geographically*, the same codebase and AI stack applies to every university city in Bangladesh — Chittagong, Rajshahi, Sylhet, Khulna — each facing identical structural rental problems. *By user segment*, the platform extends from students to young professionals and family tenants. *By monetisation*, the platform can introduce verified listing badges, premium tenant profiles with student ID verification, a bKash/Nagad in-platform payment processing fee, and anonymised area rent-trend reports for researchers and property developers.

Longer-term, a consistent on-time payment record on **Thikana** could serve as a financial history signal for students applying for formal credit products — addressing a meaningful financial-inclusion gap in Bangladesh's growing digital economy.

Thikana competes on *trust and clarity*, not listing volume.
Submitted for **Mindsparks 2026 | CodeFront Challenge** by Team IUT_TwinCoder.